Machine Learning Pipeline & Explainability Report

Comparative Evaluation: Original vs Synthetic (TSTR) vs Combined Models on Holdout Real Data

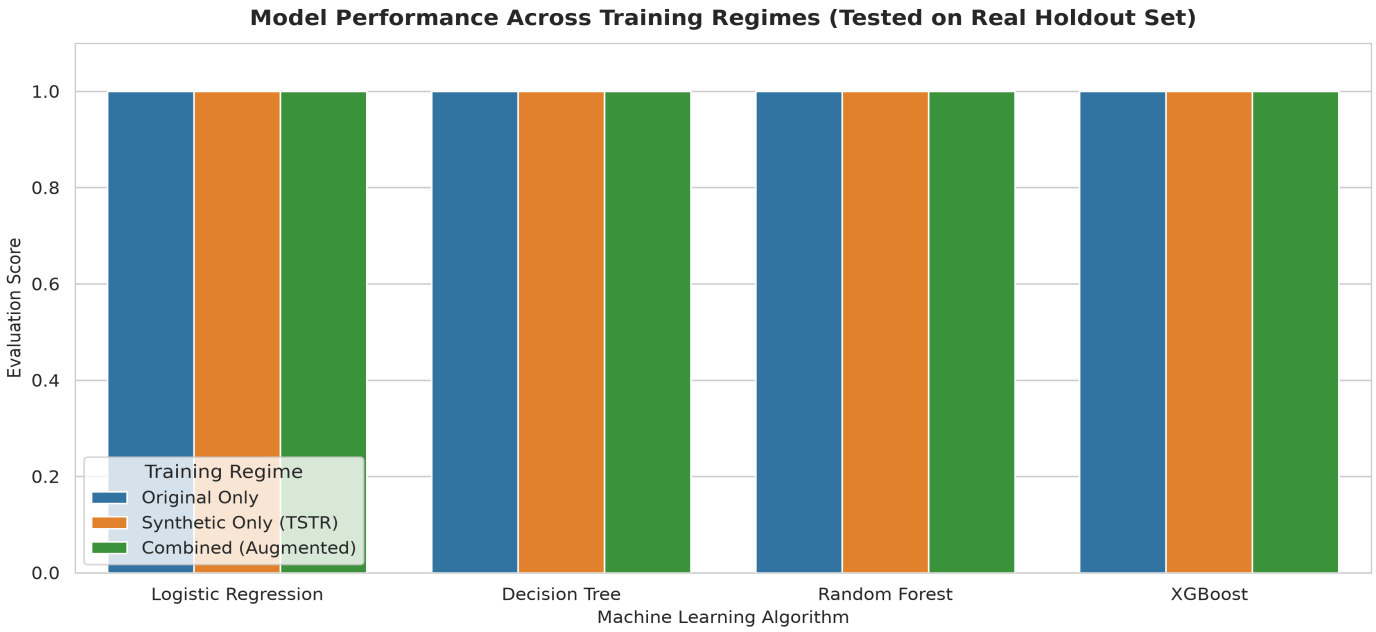
1. Machine Learning Benchmark Table

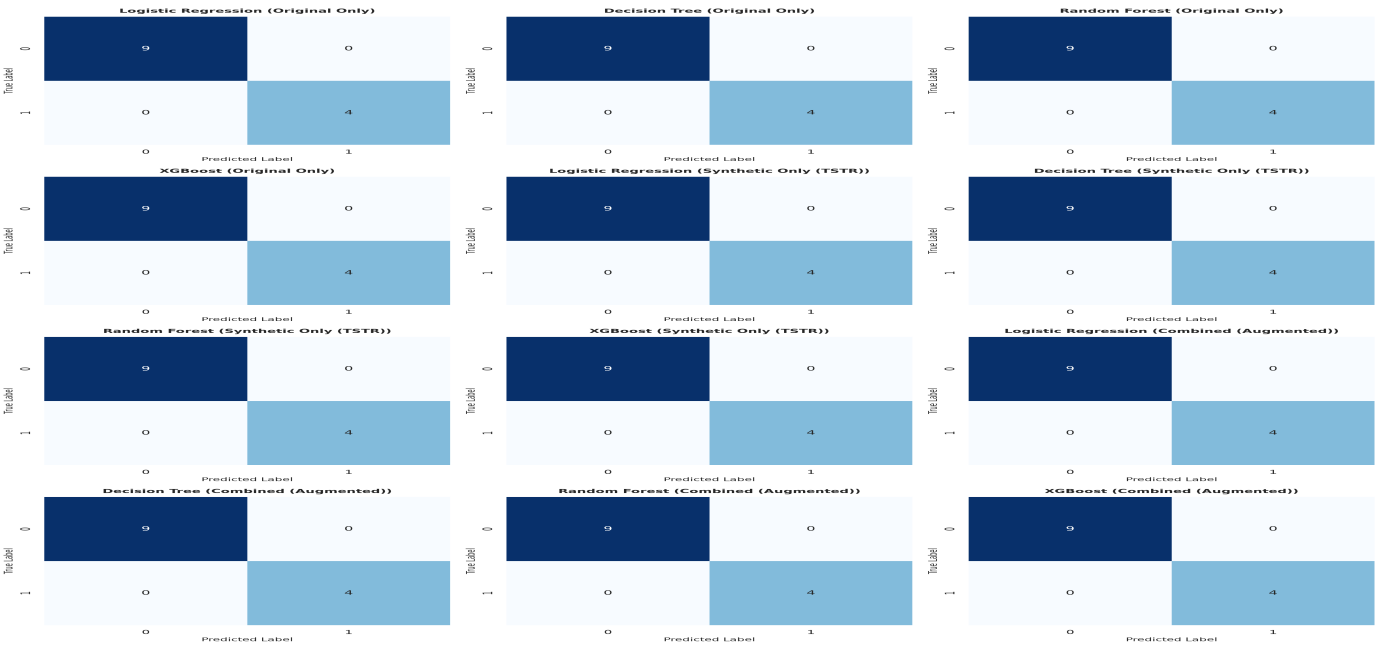
Model Algorithm	Training Regime	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	Original Only	1.0000	1.0000	1.0000	1.0000	1.0000
Decision Tree	Original Only	1.0000	1.0000	1.0000	1.0000	1.0000
Random Forest	Original Only	1.0000	1.0000	1.0000	1.0000	1.0000
XGBoost	Original Only	1.0000	1.0000	1.0000	1.0000	1.0000
Logistic Regression	Synthetic Only (TSTR)	1.0000	1.0000	1.0000	1.0000	1.0000
Decision Tree	Synthetic Only (TSTR)	1.0000	1.0000	1.0000	1.0000	1.0000
Random Forest	Synthetic Only (TSTR)	1.0000	1.0000	1.0000	1.0000	1.0000
XGBoost	Synthetic Only (TSTR)	1.0000	1.0000	1.0000	1.0000	1.0000
Logistic Regression	Combined (Augmented)	1.0000	1.0000	1.0000	1.0000	1.0000
Decision Tree	Combined (Augmented)	1.0000	1.0000	1.0000	1.0000	1.0000
Random Forest	Combined (Augmented)	1.0000	1.0000	1.0000	1.0000	1.0000
XGBoost	Combined (Augmented)	1.0000	1.0000	1.0000	1.0000	1.0000

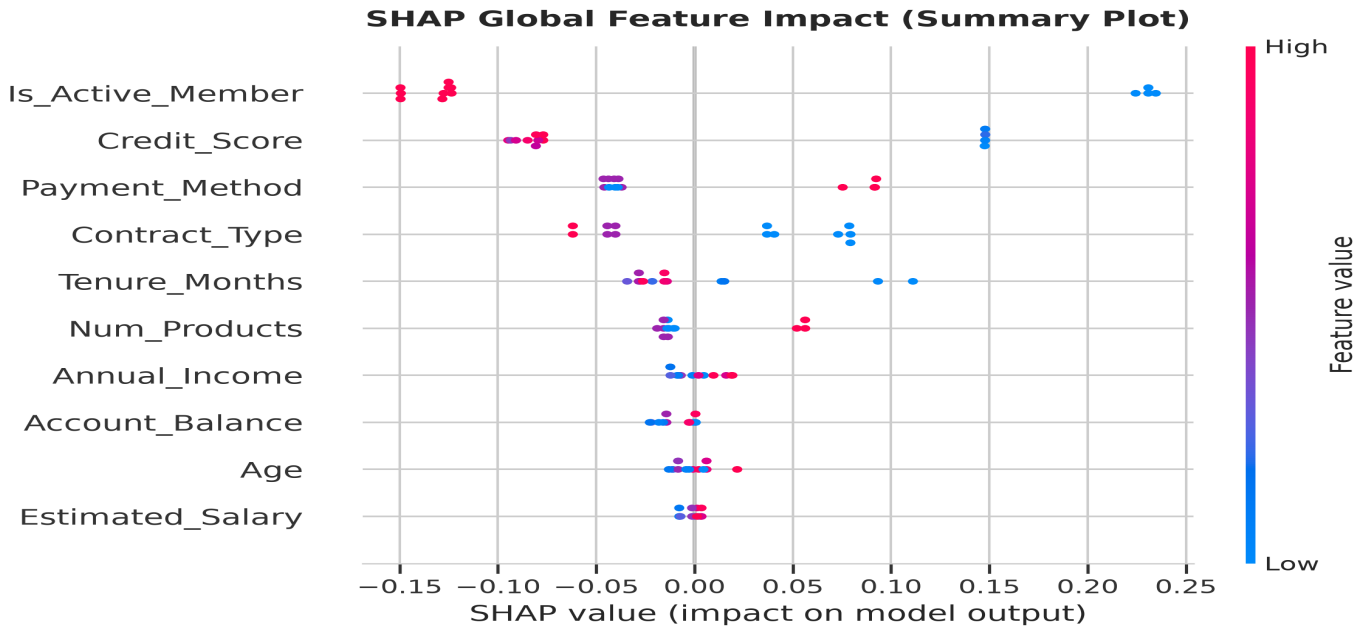
2. Experimental Methodology & TSTR Interpretation

Train on Synthetic, Test on Real (TSTR) Principle: The ultimate utility test of synthetic data is whether a model trained solely on synthetic samples can accurately predict labels on unseen real-world test data. Close alignment between Original and Synthetic metrics validates high downstream utility without compromising real user privacy.

3. Model Visualizations & Explainable AI (SHAP)


Model Performance Across Training Regimes (Tested on Real Holdout Set)
Evaluation Score
1.0
0.8
0.6
0.4
0.2
0.0
Training Regime
Original Only
Synthetic Only (TSTR)
Combined (Augmented)
Logistic Regression
Decision Tree
Random Forest
XGBoost
Machine Learning Algorithm


Logistic Regression (Original Only)
Decision Tree (Original Only)
Random Forest (Original Only)
XGBoost (Original Only)
Logistic Regression (Synthetic Only (TSTR))
Decision Tree (Synthetic Only (TSTR))
Random Forest (Synthetic Only (TSTR))
XGBoost (Synthetic Only (TSTR))
Logistic Regression (Combined (Augmented))
Decision Tree (Combined (Augmented))
Random Forest (Combined (Augmented))
XGBoost (Combined (Augmented))
True Label
Predicted Label
9
0
0
4


SHAP Global Feature Impact (Summary Plot)
Is_Active_Member
Credit_Score
Payment_Method
Contract_Type
Tenure_Months
Num_Products
Annual_Income
Account_Balance
Age
Estimated_Salary
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
0.20
0.25
SHAP value (impact on model output)
High
Feature value
Low